

Centered bumps simplify the polynomial-die proof

Companion note, 30 September 2026

This is an alternative to the evaluation-at-zero bumps in `polynomial_exceptional_die.tex`. It leaves the theorem and the finite realization unchanged, while reducing the order-response calculation to the two old dice adjacent to the distinguished die.

For fixed distinct interior real nodes u_q , choose disjoint rational neighborhoods I_q and, inside each, m rational subintervals left of u_q and m right of it. Let V_q be a neighborhood of u_q separating those two groups of subintervals. For each $t \in V_q$, solve the two interval-moment systems with target vectors

$$(1, t, t^2, \dots, t^{m-1})^T \quad \text{and} \quad -(1, t, t^2, \dots, t^{m-1})^T.$$

Their matrices are fixed invertible rational matrices. The resulting piecewise-constant function $h_{q,t}$ is continuous in t in its coefficient vector and is rational when t is rational. Uniformly for t near u_q , it is bounded. For every $g \in \mathcal{P}_{m-1}$,

$$\int_0^t h_{q,t}(x)g(x) dx = g(t), \quad \int_t^1 h_{q,t}(x)g(x) dx = -g(t).$$

Its full moments through degree $m-1$ vanish.

Use the same rational nodes t_q , disjoint index sets Q_j , and Vandermonde coefficients $\delta_{j,q}$ as in the main proof, and set

$$f_j(x) = 1 + \sum_{q \in Q_j} \delta_{j,q} h_{q,t_q}(x).$$

As $t \rightarrow u$, the coefficients δ tend to zero while all bumps remain uniformly bounded. Thus positivity follows exactly as before. The full-moment cancellation of all multiple-bump terms is unchanged.

Fix an order placing the distinguished die after k old dice. A single bump from a die before the distinguished die has kernel

$$\frac{x^a(t-x)^b(1-t)^{m-k}}{a!b!(m-k)!}, \quad a+b=k-1.$$

Evaluation at $x=t$ makes it zero unless $b=0$. Hence only the old die immediately preceding the distinguished die contributes, with polynomial

$$\frac{t^{k-1}(1-t)^{m-k}}{(k-1)!(m-k)!}.$$

Likewise, on the right side only the old die immediately following the distinguished die contributes, with negative polynomial

$$-\frac{t^k(1-t)^{m-k-1}}{k!(m-k-1)!}.$$

Since the aggregated correction functional B_j is the same B for every old die, the total correction is again

$$B \left[\left(\frac{t^k(1-t)^{m-k}}{k!(m-k)!} \right)' \right].$$

At $k = 0$ or $k = m$, only the applicable neighboring die is present. This proves the order-response identity without summing over all old positions.

Explicit kernel. The root agent and the independent-directions agent additionally obtained the following convenient rational version. Put $H_m = \sum_{j=1}^m 1/j$ and

$$c_\ell = H_m + \sum_{j=1}^{\ell} \frac{(-1)^j}{j} \binom{m}{j}, \quad 0 \leq \ell \leq m-1.$$

The function that is constant c_ℓ on $[-\ell-1, -\ell]$ integrates every polynomial of degree at most $m-1$ to its value at 0. This follows by taking an antiderivative and applying the backward Lagrange differentiation formula on nodes $0, -1, \dots, -m$. After translation and scaling to $[t-h, t]$, its constants become $(m/h)c_\ell$. Negative reflection about t supplies the right side. The bound $|c_\ell| \leq H_m + 2^m$ gives an explicit positivity certificate when the perturbation coefficients are sufficiently small.